STAFF REVIEW · H446 054 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A queue.

2. Keys to associated values.

3. Different keys map to same hash location.

4. Slot 4. All three hash to 2; 9 occupies 2, 16 occupies 3, and 23 probes to 4.

I Do — source worked steps

1. Explain the trace — Both keys collide in bucket 2; a table must resolve collision by chaining or probing.

We Do — guided CHECK

1. Explain the trace
No collision: key 8 maps bucket 3, key11 maps bucket1.

You Do — source mark schemes

1. 8→3 [2],13→3 [2],19→4 [1], 8/13 collide [1].

2. var d=new Dictionary<string,int>{{"Ada",80},{"Ben",65}}; [3]; if(d.TryGetValue("Cleo",out int mark)) ... [3], handles missing key.

3. Chaining keeps multiple entries per bucket [2]; linear probing searches next available slot [2]; load factor and clustering affect lookup [2].

Exit Ticket — source answer

1. False. Finite bucket spaces mean collisions may occur.